

AnyCam 2.6.1 Audit Report

Release audit - 27 September 2026 - Tier 1 live-feed work - tag 2.6.1

Correction, 27 September 2026. This report supersedes the 26 September 2026 revision. That revision attributed compliance rows to "Part 6 - Project Engineering" of the best-practices document. No such part exists. The checks were performed and passed; the attribution was wrong. The affected rows are relabelled below and the scope note is new. No finding changed.

Status: superseded by 2.6.2 and never installable. The image build fails on a stale ffmpeg apt pin carried since 2.6.0-rc2.3. No code in this release is at fault. The code changes below are all carried forward into 2.6.2 unaltered.

2.6.1 carries Tier 1 of the live-feed quality work: three of four planned items, plus a repair to the release gate.

Item 1 - low-latency ffmpeg flags

`_launch_snap` now always passes `-fflags +nobuffer` and `-flags low_delay`. Both are standard for live RTSP and neither affects stream detection.

The merge defect this avoided

ffmpeg accepts a single `-fflags` value. The existing code added `-fflags +discardcorrupt` conditionally for cameras with H.265+ history. A second `-fflags` argument would have silently dropped `+discardcorrupt` on exactly those cameras. The new code appends to one string.

Held back deliberately

`-probesize 32`, `-analyzeduration 0` and `-reorder_queue_size 1` sit behind a new `low_latency` option, default off. A small probe size can stop codec identification; a one-packet reorder queue removes the RTSP jitter buffer.

Item 3 - Fast Stream Start gated

Suppressed at 3840x2160 HEVC, where software decode cannot beat rpidvid warmup (2.5 to 7 seconds), so the parallel decode spent CPU on frames that never rendered and opened a second RTSP session. Width is read from the ladder's active profile, not the camera record, because a step-down may already have moved off 4K. Still applies below 4K and to h264.

Item 4 - ONVIF SOAP honours the brand cooldown

`_onvif_soap` is synchronous and opens up to two TCP connections per call, because cameras answering HTTP 400 to SOAP 1.2 get a SOAP 1.1 retry. It never consulted the throttle, so a re-auth fired 1 + N calls back to back inside the cooldown window. `_rerun_onvif_auth` now awaits `_throttle_wait_if_needed` before `GetProfiles` and before each per-profile `GetStreamUri`. Closes the latent issue in CLAUDE.md.

Release gate - Windows portability

`verify_release.py` could not run on Windows. Three `read_text()` calls omitted encoding and hit cp1252 on UTF-8 source; `ok()` and `fail()` printed U+2713 and U+2717, which cp1252 cannot encode. Both fixed. Invisible on the Pi because Linux defaults to UTF-8.

Item 2 - deferred, not implemented

Replacing per-frame polling with a persistent multipart/x-mixed-replace response was not implemented. The docstring on `handle_snapshot` records that per-frame polling was deliberate, because Home Assistant ingress nginx terminates long-lived multipart streams early. Per CLAUDE.md rule 8 that prior finding outranks a fresh assumption. The finding may be stale, since `ingress_stream: true` is set; the next step is a throwaway test, not a rewrite.

Best-practices compliance

Measured against *docs/AnyCam_Coding_Best_Practices.md*. CLAUDE.md rule 1 requires this section in every audit. It was present through 2.3.2 and dropped during the 2.4.x cycle. It is restored and stays.

Scope note. That document contains Parts 1 to 5 and a Sources Summary. It has no Part 6. CLAUDE.md describes a "Part 6 - Project Engineering" covering security, dependency management and version control, so the working agreement and the document disagree. Rows below marked *outside Parts 1-5* were checked against ordinary engineering practice, not against a written AnyCam standard.

Standard	Check applied	Result	Note
1.1 Truncated function	AST plus duplicate-def scan	PASS	140 top-level definitions, no duplicates.
1.2 Syntax errors	compile() to bytecode	PASS	Gate 1.
1.4 Missing await	Every added call reviewed	PASS	Both added <code>_throttle_wait_if_needed</code> calls are awaited.
1.4 Blocking I/O in async	Checked for <code>time.sleep</code> , <code>requests</code> , <code>open</code>	PASS	None added. The sync SOAP helper stays in <code>run_in_executor</code> .
1.4 await in loop	Flagged the new await inside for prof in profiles	NOTED	Intentional. Serialising is the point of the fix; <code>asyncio.gather</code> would defeat the cooldown. Throttle value computed once outside the loop.
1.5 Naming	PEP 8 on all new identifiers	PASS	<code>CFG_LOW_LATENCY</code> <code>UPPER_CASE</code> ; locals <code>snake_case</code> .
1.5 Line length	Measured every added line	NOTED	9 added lines over 79 chars against a 6.0 percent existing baseline. Two self-inflicted ones shortened before tagging.
1.5 Bare except	Scanned the diff	PASS	The one added handler names <code>AttributeError</code> and <code>OSError</code> .
1.5 Mutable defaults	Reviewed new signatures	PASS	No new function signatures.
Parts 2 to 4	No JS, HTML or CSS changed	PASS	Deferred Item 2 would have touched JS.
5.1 Escaping context	No f-string HTML or CSS added	PASS	Not applicable.
5.4 Sneaky errors	Diff against every row	PASS	The <code>-fflags</code> merge defect is the row this diff could have hit. Avoided by design.
Outside Parts 1-5: security	Reviewed for <code>shell=True</code> and unvalidated input	PASS	<code>create_subprocess_exec</code> takes an argument list, not a shell string. Not covered by a written AnyCam standard.

Standard	Check applied	Result	Note
Outside Parts 1-5: dependencies	Reviewed new runtime dependencies	PASS	None added. Note this check does not cover apt pin staleness, which is what made this release unbuildable.

What this audit missed

This release passed all five gates and every row above, and still could not be installed. The reason is recorded here so the gap is not repeated.

The Dockerfile pinned ffmpeg to an exact version that both Debian and the Raspberry Pi archive had already deleted. No release gate inspects the Dockerfile, and the best-practices document has no standard covering dependency pinning. The dependency row above checked only whether new dependencies were introduced, which is the wrong question.

Files changed

File	Change
camera_discovery.py	Added CFG_LOW_LATENCY. Merged +nobuffer into the single -fflags value, added -flags low_delay, probe flags behind the new option. Gated fast_stream_start at 4K HEVC. Added two throttle waits in _rerun_onvif_auth. Version to 2.6.1.
config.yaml	Added low_latency option and schema entry.
run.sh	Exported LOW_LATENCY and logged it.
translations/en.yaml	Added Low Latency Probe description; noted the 4K HEVC limit on Fast Stream Start.
verify_release.py	encoding=utf-8 on three read_text calls; UTF-8 stdout.
CHANGELOG.md	Prepended the 2.6.1 entry